

regtools: facilitating manipulation, analysis and visualization of data from Norwegian health and population registers

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Software

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Summary

The use of health and administrative registers, often in combination, is an essential component of modern epidemiological research. Among the Nordic countries, in particular, an array of registry sources offers high-quality, broad-coverage data collected across many years (Laugesen et al., 2021). The use of existing data from registers in research circumvents the data collection process, thus making research more cost and time effective (Thygesen & Ersbøll, 2014). Moreover, health and administrative registers offer enormous sample sizes and high representativeness that are much needed, particularly in epidemiological research. Although registers are rich sources of information, pre-processing and working with the large datasets they produce can be challenging and time-consuming – especially for researchers with limited programming experience – and the process is vulnerable to both unintended variations across projects and highly consequential errors.

The regtools R package is an open-source toolkit designed to aid researchers in performing efficient and well-documented manipulation, analysis and visualization of individual-level data from Norwegian health and population registers. With it, we aim to facilitate reproducible descriptive epidemiology based on Norwegian health data, supplemented with sociodemographic information, such as income and education information, from other registry sources. The package includes functions to validate, filter, and link health (diagnostic) and administrative (sociodemographic) data. For transparency, each function creates a log that documents the function's internal data processing, warnings/errors, and corresponding outputs. Finally, considering the extensive use of registers in epidemiological research, regtools includes functions intended to help users compute common descriptive epidemiology statistics, such as prevalence and incidence rates, and visualize the underlying data.

Statement of need

Due to their characteristics, Nordic registers are highly regarded for their unique potential in current epidemiological research (Jervelund & Montgomery, 2020; Maret-Ouda et al., 2017). In the last decades, epidemiological research in the Nordic countries has harnessed the advantages of registry data, such as primary and secondary health care registers (Miettunen et al., 2011; Thygesen & Ersbøll, 2014). In part, this is due to the introduction of personal identification numbers into the Nordic population-based health registers, which enables linkage to other data sources, allowing long-term, multi-dimensional follow-up of individuals in the population.

Registry data from national statistical institutes (NSIs) are a widely-used source of auxiliary information in this regard.

In Norway, the Norwegian Patient Registry (NPR) is used in a large variety of research projects (Bakken et al., 2020). As of 2025, more than 1000 research papers have been published based on data from the NPR (Norwegian Institute of Public Health, 2024). Statistics Norway (SSB) provides sociodemographic individual-level data on various topics, such as social welfare, education and income. For instance, between 2021 and 2024, SSB delivered around 900 individual-level data assignments to both public authorities and research institutes for analytic and research purposes (Statistics Norway, 2024, 2025). Despite their relatively widespread use in research, health and administrative registers are not designed with research or statistical purposes in mind. This creates numerous potential challenges, inefficiencies, and vulnerabilities in the process of carrying out epidemiological research using linked register data.

Considering the wide range of researchers using individual-level registers in Norway, it is highly likely that there are differences in the way researchers pre-process and prepare their data for analysis. Access to register individual-level data is regulated by strict confidentiality laws, which makes “hands-on” training or tutorials hard to access and standardize. The use of proprietary software to manipulate and analyze the data further hinders efforts to ensure reproducibility and transparency across research projects working with the same data (Mathur & Fox, 2023). In this context, we have identified the need for an open-source toolkit to assist researchers working with Norwegian individual-level registry data to prepare, manipulate, and analyze it in a robust, transparent, and reproducible way.

State of the field

Considering the confidential nature of health and administrative register data, a large proportion of the code used in research is not publicly available. Nonetheless, some few open-source software solutions have been developed to deal with the processing and analysis of Norwegian survey and register data. For instance, phenotools (Hannigan et al., 2021) is an R package that aims to facilitate efficient and reproducible use of survey information from Norwegian cohort data. Similarly, csverse (White, 2025) is a suite of packages focusing on the preparation and analysis of Norwegian disease surveillance and linked register data. Finally, the Norwegian statistics bureau (Statistics Norway) has developed a handful of packages addressing statistical disclosure control, retrieval of publicly available data, classification standards and survey weighting and estimation (Jentof et al., 2026; Langsrud & Bruusgaard, 2026; Langsrud & Lupp, 2025).

In summary, these existing projects have showcased the potential of using open-source R packages to assist researchers working with Norwegian survey and register data. The justification for building regtools as a new package, rather than contributing to existing packages, is that currently available software focuses on relatively narrow, well-defined use-cases (e.g. tabular suppression, disease surveillance) as opposed to broad operations with large, linked individual-level datasets aimed at facilitating descriptive epidemiological analyses. In addition, building a new package has enabled us to create a streamlined framework including only functions relevant to this research area. For instance, it integrates relevant functions originally developed by Statistics Norway to facilitate the creation of realistic synthetic data.

Software design

The package regtools was built following the principles of modularity and flexibility, as such the functions included in the package operate independently from one another, which increases their possible application in various research projects. Given the potential of multinational registry-based cohort studies (Maret-Ouda et al., 2017), it is important to note that, while the

89 package workflow is originally designed for Norwegian data sources, its flexibility may allow for
90 use with other national registries.

91 Considering that the target user base of `regtools` (social scientists, epidemiologists) might have
92 limited programming experience, we aimed to provide a user-friendly and educational framework
93 and functions. For instance, one of the first challenges researchers working with population-
94 based registers encounter is that of efficiently manipulating very large datasets into smaller
95 and tidier datasets with which they can work analytically. The `regtools` package includes
96 reading and filtering functions that support files in parquet format ([Apache Parquet, 2025](#)),
97 which seamlessly enables users to efficiently work with larger-than-memory files in R without
98 requiring deeper knowledge on the inner workings of parquet format objects. Furthermore, the
99 logs created by each function can help researchers keep track of and document all manipulation
100 or processing steps applied to their datasets. The package also includes functions that are
101 particularly useful for descriptive epidemiology analyses, such as the computation of prevalence
102 and incidence rates, along with a function for visualizing the results. There are some specific
103 challenges related to Norwegian registry data that are addressed in the helper functions of
104 `regtools`, such as harmonizing municipality codes and retrieving population counts from SSB's
105 open data.

106 In addition to helping solve practical challenges associated with processing, manipulation,
107 and analysis of Norwegian register data, `regtools` provides “hands-on” guidance on how to
108 efficiently work with individual-level registry data for epidemiological research. The functions
109 in the package are intended to serve as a loose framework that can be adapted by researchers
110 working with similar data and research questions. The package includes a series of vignettes
111 explaining the main functions and real-life examples of descriptive epidemiology. The vignettes
112 and possibility of creating synthetic individual-level datasets (`synthetic_data()`) also allow
113 research-groups to use the package as zero-risk training material for new members, and to
114 plan and structure analytic projects prior to obtaining data access.

115 Research impact statement

116 The most prominent use of the `regtools` package to date has been in providing analyses on
117 time trends in autism diagnoses in Norway over recent years, published as part of a national
118 public health report ([Martinez Sanchez et al., 2025](#)). The report received approximately
119 500,000 unique visits throughout 2025, and the analytic component based on work carried
120 out using `regtools` was the most visited section. Core features of this report, which included
121 regional breakdowns and visualization of autism incidence spanning 12 years and prevalence
122 analyses with stratification by demographic characteristics retrieved from Statistics Norway,
123 showcase the key advantages of the software. The package has additionally been used for
124 analyses of time trends in other conditions (e.g., anxiety and depression), with publication of
125 results forthcoming. Moreover, community-readiness has been demonstrated by the acceptance
126 of talks and posters presenting `regtools` at conferences in the fields of epidemiology, autism
127 research, and reproducibility in scientific publishing (e.g. [Martinez Sanchez & Hannigan, 2025](#)).

128 AI Usage Disclosure

129 No generative AI tools were used in the development of this software or the writing of this
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